

ActInf Livestream #006.2: "A tale of two densities"

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hello and welcome everyone to the active inference live stream take two with an awesome set of flexible and awesome participants this is active inference live stream 6.2 on october 20th 2020 welcome to team com everyone we are an experiment in online team communication learning and practice related to active inference you can find us on twitter at inferenceactive you can email us find us on keybase or at our youtube channel this is a recorded and an archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here and also as far as video etiquette for live stream goes please mute if you're going to have noise in your background raise your hand so that we can get to hear from everyone and use respectful speech today just as a quick point of news our friend and colleague jared tumul has released a very interesting blog post called spinning up in active inference and the free energy principle a syllabus for the curious and it is a great place for beginners as well as for those who want to go deeper into one of the various domains that are touched upon by active inference and the free energy so today in active stream 6.2 we are going to have a warm-up we're going to have take two on our intros as i kick out this other alex we are going to then come to a discussion of the paper last time in 6.1 we talked a lot about the goal of the paper and we went through the roadmap and the abstract in quite a lot of detail this time we're going to be highlighting some excellent follow-up comments from our participants we're going to be talking about pattern versus structure metaphors and more and then we'll go through the figures and make sure to talk about some of the similarities and the differences between bayesian networks and forney factor graphs great on to the intros and check-ins so for this section please just introduce yourself and your location say hello provide a little short intro if you'd like and pass it to somebody who hasn't spoken yet so i'm daniel i'm in california and i will pass it to our first time discussant kate i'm sorry i'm kate um i'm based out of the university of edinburgh i'm philosophy of cognitive fossil cognitive science student and i'm interested in lifeline continuity and trying to find a

satisfactory definition of what it is to be a living system and i will pass to mel i think mel is not joined in yet let's go to shannon i think mel's here i don't know yeah absolutely mel started uh mel started her intro actually so mel is absolutely here we can hear you i don't know if daniel can hear you yeah weirdly i don't hear you mel could you just reload sorry and shannon go ahead okay no i don't think he sees you this is a new and fun kind of just reload the page mill like just just click refresh yeah exactly perfect i see that mel take it away thank you there you go we're good yes yes beautiful all right cool i'm now andrew i'm in ohio i'm a doctoral student in philosophy of biology and company science i'm interested in the same question as kate and i will pass to alex kiefer uh hi i'm alex kiefer um i'm uh in the philosophy department at monash university currently in new york city um i guess i'll probably be wind up defending structural representationalism about generative models today i don't know i'll pass it to maxwell i guess i'll probably be agreeing with you in the end uh i think alex i'm maxwell ramstead uh i'm a postdoctoral fellow at mcgill university in montreal i'm speaking to you from montreal right now i work on multi-scale active inference and i'll pass it to i guess ale you haven't spoken yet you're right bye we are three alex um well hello everybody um i'm ali or alejandra um i'm a professor in the department of psychology specifically in the department of um cognitive sciences and well i'm in also the same question that you were talking about what exactly is to be alive and maybe going further what is cognition and perception so yeah um that's me oh well i'm i'm in mexico um so i'm going to pass it to um sasha hi i'm sasha i'm based out of davis california and i'm a neuroscience graduate student um studying kind of the molecular and developmental aspects of the brain and i will pass it to shannon hi i'm shannon i'm based out of uc merced in california but i'm in south dakota right now i'm also interested in how active infrareds can help us learn about the brain and music cognition um as well as how we can learn about how humans coordinate in music or other sort of group interactions and i will pass it to the other alex hello uh i'm alex watkin i'm a researcher at system management school in moscow russia i'm trying to find possible ways of integration active inference with system engineering and system management frameworks so who's else cool i think that is all of us and thanks everyone for bearing with these funny technical glitches i'm also recording this locally so worst case we'll just be able to have this uploaded let's get on to the warm-up questions so for the warm-up questions just anyone who wants to speak would be awesome to hear from you just to sort of get started what is something that you've been working on or learning about recently whether it's directly linked or just something that influences your mind state coming into today's discussion and you can popcorn in and also i'll be watching the stack

if people raise their hand i'll start while people are raising their hand we just had complexity weekend this past weekend which was a really exciting and interesting time about 100 people from 37 countries and forming cool teams on different topics really learned a lot mel hello yep that worked great beautiful um yeah so recently so i've been i've been sort of puzzling for a couple years now god it's been that one um over how to conceptualize on the free energy principle um and so that's pulled me into work in philosophy of science on models and modeling so so all different kinds of ways of conceiving of formal models and what they do and then kind of parallels in uh in biology in behavioral ecology and ethology and evolutionary biology with with models that sort of [Music] having some sort of empirical concept cool anyone else want to share something they're working on or i can also just raise the second warm-up question just in case people want to answer that one which is what is something that you'd like to have resolved by the end of today's discussion whether it's a thought about the paper specifically or something general um sure fellow gitzer i'm not sure that is shannon go ahead hey um i wonder whether the uh like structural representation that we might be defending later today um whether that structure could just be a set of attractor dynamics like is that enough to be a structure um so i think alex might be raising his hand to have something to say yep alex oh yeah i mean i was gonna speak anyway i mean i would say yes to that question probably um it's not just my opinion but um i was going to answer both questions i guess so um apart from trying to figure out the free energy principles still uh lots of conversations with people i've been trying to um i've been working up to trying to implement a variational autoencoder to figure out how that actually works so um been doing a lot of sort of more technical work um something i'd like to have resolved is just to figure out exactly where um in activism or the version of an activism that's defended in this paper is inconsistent with representationalism if it is um so like it's i don't need there to be residual disagreement but i just want to figure it out ooh sounds good any last thoughts on these warm-ups before we jump into it we have a lot of very interesting slides kate go for it yeah i guess the biggest thing that i want to resolve by the end of the session is i'm still not really clear on the status of the generative model because at some points it sounds as if that's something that's almost causal that it's controlling the animals behavior or the organism or the system's behavior and at other points it's a model it sounds more like it's a model that we use to describe the systems dynamics rather than something that is including beliefs or being causal and that's something i'm still not entirely clear on so i think that's a big thing i won't resolve cool all really interesting topics thanks for sharing and we're gonna jump into it because a lot of these are on

the slides ahead so the paper that we're gonna be coming back to today is a tale of two densities active inference is an active inference by ramstead kirchoff and fristen and just to rehearse the goal of this paper they aimed to clarify how best to interpret some of the central constructs that underwrite the free energy principle and its corollary active inference in theoretical neuroscience and biology namely the role that the generative models and the recognition densities play in this theory aiming to unify life and mind and so last week we talked about what these two densities are and we talked about the tail that integrates the densities we also walk through the road map how they build from a to z by talking about statistical models and representations introducing active inference and then combining that with an active inference in this sort of inactivism 2.0 framework before building towards multi-disciplinary research heuristics for cognitive science so that was last week let's get into the new and exciting parts so this was one of the feedback comments from a participant last week and what they wrote was a key learning goal for me in building my understanding of active inference is to develop intuitive examples for the critical concepts that can act as the basis for abstraction later on it seems to me like there is probably something like a logical sequence in which this should be done but i don't necessarily always know which concepts and ideas from active inference to prioritize as an example coming from a cognitive linguistics background i tend to think of beliefs as either causal reasoning or the underlying sensory motor simulations that connect to linguistic concepts i've been trying to find new intuitive examples that can help me to understand beliefs as probability distributions as clearly this is the key for making progress in understanding active inference so my question for all of you is what is a system or a metaphor that helps you understand active inference and why and while people are raising their hand on the bottom we have just a few of the systems that we've been discussing over the last weeks we have a brain the technical and the artistic parts of the brain we have insect colonies we have dance and embodiment and movement and then on the right i guess it reflects something like the economy or changes in dynamical systems but what is that helps you understand through specifics something about active inference any thoughts on this i mean i hate to be that guy but like i started understanding active inference when i went into the math uh like it's it's a formal theory and i i think we can actually be misled by uh metaphor uh yeah uh some of my some of my reservations about the philosophical work that's been done on active inference is that it it operates on the basis of you know uh kind of unformal intuitive conceptual understandings rather than trying to see what the math that says so i hate to be that guy though no that's perfect we'll go to alex bjotkin and then to kate go ahead alex yeah for me

most important changed my views was taking seriously approach from like for process and understanding concepts about states and steady states of possible processes and the later on all this concept with a markov blanket as a border and underlying structure like generative models in links with cybernetics it's bringing me some kind of top-level picture and now i'm trying to focus on different parts to have more detailed understanding cool kate and then mel yeah i was also going to say cybernetics i think i only really started to understand what was going on once i started seeing the links with like ashby and account of survival as stability um that was when things started to make sense for me um and there's also just like a nice metaphor i quite liked and i think it's really it's a paper by ellie rudenberg i think with the union kingston rightfeld with the crooked scientist metaphor the way the scientist is trying to obtain the result that it's looking for so it has to have some sort of latch on the world in order to bring about that result but it's still trying to obtain the result that it wants in the first place rather than passively find out what will happen i thought it was quite a nice metaphor so do you think they are helpful cool so i remember i'm i'm remembering um my first sort of aha moment with the fpp and active entrance um i was i was explaining this stuff to my friend who works on on similar things but in a very different framework um in the mit media lab and we had like papers strewn everywhere and scribbling all of them and my realization in explaining this framework to him was that um that there is fundamentally at the at the lowest level at the at the level of the simplest sort of systems or at their kind of lowest scale right um no difference between belief updating and action so a a change in the system is a change in the system right and that is both a belief update and an action right um and once you build in further levels of of hierarchical organization and complexity you get this strong differentiation between them but at the lowest level of this the same thing cool and kate did you know that's a probability okay had you raised your hand there and also i'm just going to yeah alex kiefer go ahead oh um yeah i mean i was gonna that's i like that point that at least at the lowest level we'll leave updating an action or kind of same thing i guess um that's one reason that's one reason that i'm skeptical that we need to explicitly i think implicitly more traditional bayesian brain stories kind of are built on that claim as well but um i was going to say what got me into what got me into active inference also sort of a technical point was just that you really can't you can't optimize the thing you can't minimize surprisal just by ingesting adjusting and at least not okay the surprise depends in part on the sensory observations right so you can't minimize that just by changing your generative model um you have to change the observations so that's that's the technical thing that got me into it um cool and yeah thanks everyone for being flexible and

fun on the tech maxwell i really agree that the metaphor as always with metaphors it's a question about compression of semantics and meaning and at the end of the day we're talking about something specific and so we can think about different systems that help us understand how active inference is deployed in the real world but often metaphors will take off take us off the mark uh because it removes us from the real underlying details so let's return and the details are really where the devil's at uh in this case uh i think but yep i i agree we're hopefully in the end with the figures we're gonna be able to look at those details really specifically and i think that the way that figure two three and four lead into each other helped me understand a lot about the system so just to take another look at these cities the two cities that are being linked the densities on the left we have the bayesian structural representationalist and alex freely added any details here but bayesian models move between the data parameters the observations and hyper parameters or higher order parameters and from the data to the hyper parameters we call it a recognition model and to go from hyper parameters to a set of data or observations that is a generative model and the outcome is a statistical convergence of a multi-level model that represents structures of the world for example through an expectation maximization scheme we can contrast this with the inactivist school of thought which has linked rather than data and hyper data but rather has focused on the world and agents and how they're linked through perception and action so a lot more adjacencies to areas like niche construction and ecological psychology and we remember hearing from maxwell last week about how it was the desire to mathematically rigorize some of the inactivism that led him to the bayesian approach and let's now think about how these two densities are linked with a little bit more feedback from one of our participants they wrote the domino metaphor finally makes sense the physical dominoes are the physical state of the system corresponding to the recognition model and the falling of the dominoes is like the process or the dynamics of the system reflecting the generative model so here we've kind of combined these two different ways of thinking and we're looking at the relationship of the world and the agent or the system and the system surroundings and directly went ahead and combined perception with the recognition model and action with the generative model and then also just to combine that with what kate was just talking about with cybernetics this is a quote from the paper that says on this view active inference can be read as a new take on the good regulator theorem proposed by control and rosh ashby in 1970. active inference tells us about the relation between a control system the generative model with priors over action policies and a system being controlled the organism and its adaptive behavior the actual actions undertaken in and

part of the world so what what exactly does this clarify about active inference or what do people think about this metaphor um or at least way to think about how if not necessarily a metaphor how would we apply active inference to this system what are we getting at here with these similarities and differences between the two models sure mel well so the good regulator theorem the idea is that um any good control structure for some larger system has to be a model of that system it has to contain in it all of the the key variables for that system and basically what this says to me about active inference is that um in order for a system in an environment to be able to adequately react to whatever kinds of perturbations are happening in the environment it has to contain all of the key existence i call them existential variables the the variables in the environment that would lead to the system continuing to exist or not continuing to place cool and so how exactly i'm just curious um does the system come to embody all the key variables of the outside world like don't the models of the organism reflect a simplified version of the outside world otherwise there kind of is a abortion story about the map that represents the territory exactly so how does a good regulator arise in the context of an environment like that um shannon and then maxwell well i don't think the key is to be an exact like imitative model of the world or an exact map even of the world but just a model that is good enough to enable you to act and adjust in the world cool maxwell and then mel yeah i mean so formally speaking there's a difference between external states and hidden states um and i mean when everything is going well uh the external states that are modeled by the system coincide with the hidden states that are actually out there um and this is more of this it's not i'm not i'm not sure if it's guaranteed but it is strongly implied by the fact that we're minimizing uh variational free energy uh that you know if if we're if we're not generating a lot of free energy then uh the external beliefs that we have about the environment tend to reflect uh you know the causal structure of that environment but i mean manuel baltieri and chris buckley have done some interesting work showing that actually the there isn't a necessary kind of entailment relation um you can have uh you can have the the system can be acting that don't actually exist and i mean this makes sense also just evolutionarily uh perceptual systems don't track truth they enable adaptive uh behavioral loops so if you're a prey item like a bunny it's adaptive to generate a lot of false positives uh compared to you know allowing for false negatives and in terms of like predator detection it's it's more advantageous to you know make a few less costly mistakes than make a big one so um i mean yellow brinenberg and eric rivell make make that point in the uh paper that kate was mentioning just earlier but uh yeah the it doesn't actually have to hook up to anything in the real world though usually it will end up

doing so cool mel and sasha yeah so the the key term in what i've said is is key actually um so the key variables right that's that's just what keeps the system alive can imagine that we first get a system on the scene that needs just basically one parameter in its environment to be correct in order to continue to exist as and then we build up from there to systems that are progressively more and more complex and need progressively more and more complex sort of environmental scenarios to survive and then they themselves in order to survive in those more complex regimes need to build in complexity right in in terms of their the of the world that they enact right and i think that says something interesting about um how we conceptualize uh cognition and the onset of cognitive um as sort of um and this is peter garcia smith's line on the subject is is it's a response to a complexifying environmental circumstance right cool sasha then alex keefer um yeah uh one key phrase that really helped me better understand um active inference um is uh good enough and that's what really um kind of put it all together that um while we're going about and reducing uncertainty about our system it just has to be good enough to make the next action or in the evolutionary sense to survive and that metaphor has really helped me uh go through all this um all these high level concepts so thank you for mentioning that alex kieffer then alex fiatkin thanks um yeah i just wanted to just note that there's there's you don't need to contrast accuracy or you know answerable to the truth-ness with um enabling adaptive behavior uh as as like a dichotomy um i'm not sure if that's what was intended or not but this one reason i've had trouble getting into understanding the uh sort of anti-representationalist viewpoint is that um you know structural similarity is a matter of degree accuracy is a matter of degree in this sense so um i mean yes your hidden state representation doesn't have to map completely accurately onto reality doesn't have to capture every detail but if it doesn't do that to any degree um you're screwed so you know um maybe we need systematically um sort of simpler or biased representations in order to do the job but i think there's still a implied relationship to the truth there cool alex yeah thanks uh i want to change the system's level of consideration of possible application of active inference and for me interesting on a personal level in terms of day phenomenology and especially professional and or a working day phenomenology how it works what is a generative model and like an example and what i won't discuss if i get it correctly or how it's good possible develop to think about it but if for example some doctor he know different disciplines and with different ontologies and when he met a patient he started his actions by his generative model which activates with recognition models to serve as a doctor depends on exact situation and exact case and if it's so from for example from another level from team level if person uh behave

professionally in some discipline it became like a perception model for a generative model of the team and what does it mean in terms of learning and education of team members because if we can how some in some way to link recognition models with ontologies and starting to work with it more again cybernetically it's possible could be very interesting at least for me cool agreed the the ontologies that how we think about the world definitely influences our perception and sasha do you have something else otherwise okay shannon go ahead um i was just going back to alex keefer's point that your model like there has to be some truth relationship with the world and if we're going to the dominoes and the process needs to be that the fall to make some cool pattern if there's an interruption like so now this domino can't reach the next domino like that's akin to a disruption and how truthy the model is of the world and it disrupts the action that you're able to take or it disrupts the um you die instead of being able to because your model isn't accurate at that stage for the processes to continue happening to do the next action or to survive cool cool discussion on this domino system let's talk about counterfactuals and also return to some of the great questions alejandra raised in previous weeks this is from another participant's feedback they wrote what i'd like to clarify if possible is the following am i right in understanding the generative model as a control system that is dynamically instantiated between an organism and an affordance in the context and the recognition density as a relation between the sensory inputs and the possibilities for action policies encoded in the organism if so how does this work when action is instantiated by counterfactual forms of cognition or simulation i.e those that do not rely on affordances in the current external context and so on the bottom right there's a a younger person looking into the mirror or maybe vice versa and seeing an older person and this is reflecting how not only is there a deep temporal aspect to our action in the current moment meaning that there's often affordances that aren't available to us at the second that we're actually making the decision but also that we think about future possible affordances that don't exist based upon the sensory data that we're getting or do they and then on the right side i have a few bacteria because this is like an example of a system that we might not think of as having a simulation or counterfactual based mechanism of action often bacterial behavior can be summarized quite concisely by just the short term gradients that it's ascending or descending so for anyone to raise their hand and chime in how how do counterfactuals play into this and how do we make sense of affordances that aren't in a current ecosystem but are still important for behavior maxwell well i just want to comment on the first part of that question um so i mean i think it's a bit simpler than that in terms of the relationship between the recognition and the generative

models or densities um so your recognition model is just your post your posterior the all of your posteriors basically all of the all of your posterior estimates over states and precisions it's like my best guess as to you know what the value of all these states are um and i mean you can think of that as uh the system's current physical state so under under the active inference um formalism the current physical state of the organism encodes uh basically the the specific parameters of these uh beliefs over external states the the actual physical states of the system encode these parameters which so yeah you embody a best guess the the generative model is the point of reference for the free energy gradients so you can think of one as like you know uh the inference model and the other is the control model in the sense that the inference model or the recognition density is sort of like my best guess right now and the generative model is is essentially you know a model of what my phenotypic preferences are and uh it's the tandem between the two that kind of gets you behavior as in um the generative model provides the point uh the point of reference uh for the definition of the variational free energy gradients i.e how close am i um to my you know desired sensory distribution and the recognition model tells you where you are effectively so it's like where where i am where i want to be and the free energy is sort of the difference between the two effectively um so i mean the first part of that question yes you're right in control system um the recognition density is just the best guess it's the posterior uh so i mean it yes it is harnessing these relations that are being described here uh but that's precisely because it's a posterior estimate of uh you know states and precisions and all that cool and one more thought on that is for an agent who knows about for example how hammers and nails are related through their internal models they might look at hammers and nails and pieces of wood and think you know later on i could build a house with this and so it's actually a future affordance but it's based upon sensory input at the current moment um cool i'm gonna just move on to the next slide and just a short little quote by buckminster fuller because sometimes i just have to he wrote rope may not be much like water not is like the wave and this is from the synergetics framework and it's getting at this notion that there's a difference between structural integrity which is like the rope and the water which are quite different they're both static and stable to pattern integrity and knots and waves can travel through a medium and so by virtue of traveling through medium kind of like that domino wave there's something similar about these traveling waves that goes beyond the mere structural integrity of their media and so i just a thought question for later is really is active inference a structural integrity or a pattern integrity i would say that it's a pattern integrity and just bringing up this notion that many far from equilibrium systems though their mechanisms

are extremely different for example the bacteria and the person there's going to be similarities because of the way that they're needing to persist from a cybernetic perspective enough let's get to that question about structural aspects of attractors so here a participant provided feedback and wrote why can't the recognition and generative models be considered as structural representations so they're coming from a perspective they wrote in which case the dynamics of the structure are taken into account as part of the representation even if we're appealing to dynamical systems the states of any particular dynamical system over time make up the structural take the lorenz attractor the lorenz attractor has a specific approximate structure around two attractor states but the states and the trajectory of any point are sensitive to initial conditions so any two initial points will lead to different precise physical states and dynamics but will still travel through a very similar trajectory which hovers around these attractor basins the states and the dynamics of a set of points over time can be taken as a whole and be identified as embodying structural representations of the lorenz system so this is something from the realm of chaos and complexity theory and the lorenz attractor is an attractor that describes a simplified system of equations describing the flow of fluid and it's a fluid of uniform depth and there are also parameters for the imposed temperature difference the gravity parameters the buoyancy the thermal diffusivity and the kinematic viscosity and the simple physical analogy here is like heating up a pot of water there's heat coming from the bottom and there's cooling coming from the top and there's this churning happening and it goes around these two different attractor states so in this case maybe alex uh or anyone else who raises their hand what is structure and what is representation and if we agree on what this system is then what is this representation debate that's happening mel unmuted mel or maxwell go ahead all right okay let's go to alex yeah mel if you want to speak english it's raised then we'll go to alex keefe and they'll raise their hand okay well i was gonna say hey let me talk um i don't have much to say about the lorenz attractor um it looks like a i mean what's i'm not sure what's supposed to be represented in this case like it i have an intuition in the case of dealing with like a biological system because you know we're intuitively representing some kind of environment i'm just not sure i'm not sure how to begin with this example um right we've got the state space and this nice graphical representation of the states of this of the state space unless we interpret this entire system as representing something i don't know what to say about its sort of content okay cool mel i don't know what a recommendation is and so i for that reason i guess i think that the question of what a representation is and what in the world is a representation is is maybe too ambitious only because i haven't solved it um but there's a

related question that's less ambitious which is what is the difference between a biological function and a representation um and that i think is more manageable in some respects in phil bio mind cogsi we um speak of the disjunction problem so we speak of and i know maxwell like hates this framing and thinks it's updated and we don't need it anymore the idea is that in order for representation to be a representation it needs to have the capacity to represent and we have a similar idea in biological function which is that in order for a function to be a function it needs to have the capacity to malfunction right so you need malfunction dab function you need misrepresentation that representation um but how do we differentiate a function from representation good question maxwell then alex cool um well so regarding the question what is the representation we targeted a very specific account of i mean in the philosophy of cognitive science a representation is basically an internal thing within an organism that stands for some feature of the environment or the organism's own body in a way that the organism can leverage to you know do something interesting we focus more specifically on the claim that generative models are structural so uh i mean this has been an account that's been uh you know opium o'brien first worked on this in the mid 2000s uh and then gladsyevsky and milkowski in the context of predictive coding and then alex kieffer and jakob howie uh you know built on that i think in a very interesting way and you know from that point of view structural representation is an internal structure that's internal to an organism that gets its representational content uh because it stands in a relation of structural similarity to some target domain in the sense that uh second order structural features of that target domain like the statistical properties of the domain are recapitulated or like mirrored in uh the the actual properties of the the representations themselves that's the first uh there's a structural similarity first property second property it's not just that these structures are in the organism they have to be exploitable in some sense so the organism needs to be able to use uh the content uh encoded in uh the representation to guide intelligent adaptive behavior um and then there are these two other uh points that i think are pretty minor it has to be detachable meaning that the system can use it offline uh like as in uh you know not not like actually engaging with the environment and it has to be able to afford representational error detection uh in a matter similar to maps um so those are the main features of structural representations according to the philosophical account on offer and uh in this paper we propose that generative models are not structural uh precisely because they don't really need these the these properties don't seem to be true of uh the generative models um i mean yeah it i i think alex and i are gonna still disagree about this even after talking about it for like dozens of hours at this point but um yeah

uh the main the main reason why uh in this paper we argue that they're not is that the generative model isn't encoded in anything uh and this is like one of the main differences between more traditional bayesian brain architectures and active inference as dan was pointing out really early on uh in traditional architectures well the recognition and generative models are just inverse the ones of the others the the recognition model is a mapping from the data to states and uh the generative uh model is a mapping from states to data so it's really just the same kind of set of connections and everything but it's just like which direction are you considering it like from the top down flow or from the from the uh the top down floor the bottom up flow of information um in the active inference scheme it doesn't work like that uh the generative model is nowhere in the dynamics sorry is nowhere in the physical organism it's only in the dynamics so uh it's very much like these dominoes falling over uh the wave of falling dominoes isn't present in any of the kind of single dominoes falling over uh the wave is is part of the dynamic you know phenomenon uh and it's it's the same with the the variational free energy and the generative model the generative model just is not there any at any at any time slice uh it only exists in so far as we consider a lapse of time as the point of reference for the free energy gradients um so from that point of view it can't possibly be a structural representation because it's not encoded by anything cool well that's that's the point a lot there let's go to alejandra melvin okay we'll go to alex alejandra i figured i figured you'd have something there yeah well i mean that was a really a nice explanation of what structural representationalism is so uh i don't have much to argue about there and in general i don't i don't want to argue more than is necessary you know that about me but um i think um the reason i want to insist on some of this stuff is just to for my own basic sanity to just feel like i have an understanding of what's going on with this stuff right um so the reasons despite i understand i think i understand exactly why from an active inference perspective um you wouldn't want to say that the generative model is a is encoded let me just say at first i think the question of whether or how it's encoded is maybe a slightly distinct question from whether it's a representation and the reason i think that it has to be a representation well there are many reasons here one argument the recognition model is an approximation to the generative model right and so that's one example of like a rational or probabilistic relationship between these models and i don't think that makes any sense if the generative model is not a representation also um so that's how that works on directive i i agree that in in a more traditional bayesian brain sense the recognition model is trying to approximate uh the generative model i mean that's that's essentially how it's working um it's an approximate posterior approximation yeah

it's an approximate posterior but it isn't approximating the generative model well it's approximating the posterior under the generative model right so so right and both both the generative and the recognition models are concerned their distributions in part over uh i have to be careful here not the actual external states necessarily but over hidden hidden states i don't how on earth you get those into the picture if it's not a representation that you're talking about because the states are environmental states they're not necessarily right we have to be as you point out in the paper we have to be careful to distinguish the generative model from the generative process we can't just identify what actually happens with the generative model we i think we need to have this is one of the fundamental questions that i think an activism often is faced with is how do we deal with yeah i mean mel brought up misrepresentation um so the fact that the fact the point the the way that you see the world the organism or creature system sees the world is not necessarily the way it is but we still have that distribution over external states as essentially part of the generative model let me just say one more thing here about encoding so the reason i think if you look under the hood a little bit so the generative model in under active inference as i understand it can roughly be identified with the non-equilibrium steady state density right something like that um so i i mean i would just say right the the dynamics sort of instantiate the generative model i think that's a really cool point and i actually think that structural representationalists didn't do quite enough to emphasize the fact that the generative model is used as a control system but i think if you look under the hood there will be some features of the system like stable structural features in virtue of which it has that nest density so in fact you will find some stable thing that you could think of as encoding the generative model maybe at least that's why i still have you know i'm still attached to my views on this cool let's do alejandra and then mel yeah um i think i i um i agree with alex or where are i'm kind of confused how the generative model can be conceived um just like this an active process um really i was like reading the paper all over again and um i actually don't get it um for me it was um the recognition density is the inversion of the generative model so uh if it is not encoded like anywhere it's i i feel kind of uh lost there um for me yeah these top-down connections this is the generative model talking about the brain specifically and the recognition density is related with this bottom up connection so you can recognize um your best guess right so if it is not encoded um what can be said about these top-down um i don't know maybe i can jump in here and just provide some points of clarification sure about the generation wall if that's okay yep um okay so uh is screen sharing activated um you could share your screen if you yeah you probably should be able okay let me try this okay so this is

from a paper called uh the graphical brain and this is what the generative models look like all right can everyone see my screen yep also we're going to be returning look at this in the center yeah so the generative model is defined as a joint probability density over all of the variables of interest in the system right so uh in this case uh the variables of interest are the states that we're trying to infer which are here s the policies that i'm trying to pursue which is here π_i and the data that i'm observing at any given time step uh which is here denoted $o_{t:m}$ just so everyone uh is clear time flows from left to right so this is the first state this is the second state this is the third state and essentially the model you only have access to this data point these are counterfactual data points uh and the whole thing is updated basically at every time step $u_{t:m}$ and so look at the generative model itself like i said is just a joint probability density it's like the probability of all these variables connected with ands effectively and when we say that the generative model is not present in the system i mean that this density here this joint probability density is never encoded anywhere in the system uh because the generative model itself is factorized so basically you take this joint probability density and then using some uh you know bayes rule and and other uh you know chain rule etc uh manipulations you can write this joint density as the product of a bunch of uh likelihoods and priors and it's these likelihoods and priors are uh updated constantly as part of the recognition density so this this prior uh you know about your your data given your states and this other prior about your states given the next data and the policy and all this these are updated uh dynamically and their current value uh like all of them together can collectively comprise the generative model uh yeah that's right sorry the recognition model the uh generative model is how these quantities are all connected the ones to the others so like if you want to think about it heuristically uh you know this uh all of these specific parameters like the s the π_i the o all that like there all of those values together as they're being updated uh comprise the uh density and the generative density or the generative model is really just these relations between uh the different quantities as they change so it's it's literally that the $u_{t:m}$ the inference process itself is what holds all these quantities together and the generative model is just a description of how those quantities flow together uh yeah and so again it's like the domino's uh falling over thing uh you know the the wave of dominoes only exists in that motion and and similarly the generative model only exists in that kind of coordinated inference or update dynamics of the quantities that are part of the recognition density $u_{t:m}$ yeah and the reason it's called the generative model at all is just because it's borrowed it's a terminology borrowed from machine learning uh in machine learning the the joint probability density over all of your variables is known as a generative

model so it's it's called that way just because that's the that's the technical term uh it's it's also called that way because if you're starting from a joint density over all of your variables you can actually generate fictive data that you would expect under this configuration um of different uh parameters oh thanks for that clarification maxwell will have those middles we'll have mel and then we'll carry on with that all sounds pretty good to me go for it mel i think kate and mill have yeah yeah okay kate and then mel oh i just wanted to um follow up to what maxell was saying about structural representation can you hear me yep go for it can people hear me yes yes yes we can hear it beautiful awesome yeah just to follow up to what maxwell was saying about um structural representations uh the the sort of detachability of the representation right that's that's what should get us the distinction between a biological function and at least a structural representation right because my feet and legs and knees are a representation of ground in some very minimal sense in the same way that fish fins are representations of of fish sins and and bird wings or representations of fluid dynamics right um but this isn't detachable from immediate environmental circumstance for offline use right um so that's what should get us the difference between uh function representation but i'm i'm curious to see what role that plays in in the ftp in active interest um if any do we want to do we want to is there is there merit and retaining a distinction between function eruption patients under under active infants in the sap or are they just sort of continuous with one another nice very good question kate did you have something there okay okay i'm gonna just continue with the slides because i'm not sure if i think kate's up next yeah oh sorry i wasn't first just wanted to apply um i think i'm still with the in the case of the the paper was really useful for me in like helping me understand what the generation model was in the fep almost all the way like when it's described as like a statistical description of the generative process the way you've explained it all of that makes sense to me um but then there's also parts in the paper where it says where you think it's like the generative model sort of um is what the organism expects and it guides what the organism does which makes it sound like the generative model is playing this kind of causal role not just for the generative model itself um and i find that kind of confusing if the given the generative model isn't something encoded by the organism um or by the system um that i'm still i get the idea of it as a statistical description of the dynamics that sounds right i just don't understand how in that how if that's all it is that it's something the organism uses it seems like it's something we use to describe the organism rather than something organism itself is in any way using um that's a good point i i've proposed in a recent paper uh that came out in entropy not too long ago that uh both are correct so the you basically

the the correct way to interpret the fep is as instrumentalism next nested within instrumentalism meaning that we can have a model theoretic uh philosophy of science reading of the fep and you know for that matter any other theory of cognition as this is a model that we as scientists are using to explain the behavior of organisms what i've been trying to argue is that the fep itself at the theory level is also saying the organism is exploiting the statistical structure of its body and movement aka the generative model to guide adaptive behavior so the model really is these harnessed relations between the variables that have to obtain for survival to persist uh and you know so it's it's sort of this idea that once the system kind of gets moving then there's kind of an inertial dynamic thing kind of keeping it moving so like you know my core body temperature is 36 and a half degrees and the fact that it is keeps a bunch of other processes in play that in turn keep me preserving my temperature uh you know by initiating adaptive behavior like putting on a parka and it gets cold and so on so uh yeah i mean both both kinds of generative models are used separately in dynamical causal modeling uh of the kind that the Friston group used to model the spread of covid we're not assuming that the process that we're modeling is itself an active inference agent um and it so you can do that but you can also make the additional assumption that it is and then you're in the game of active inference proper cool thanks max i'll go ahead kate can i come back to that go ahead go ahead yes absolutely i i the thing the kind of worry i get there then it sort of seems like you're losing the distinction between the vehicle of the model and the target of the model such that i worry it sounds a bit like just saying everything is a model of itself essentially um i mean effectively that is what we're saying and that's where it gets a little hazy for me like you know that's probably the point of disagreement between alex and i also i want to say the generative model just is the organism in the sense that like this this specific you know statistical structure that harnesses all of these different you know existential variables as mel was saying like this really just is the organism like this kind of statistical structure that keeps kind of reiterating itself and coming into existence that just is the organism and that's one of the reasons why i don't want to say that this model is a representation of anything you could just call it you know the phenotype we call it a generative model because formally speaking the same construct is borrowed from a field where in which it's called a generative model but you can think of it as the organism's phenotype and precisely because it's a bit weird to think of the model as modeling itself i kind of resist that interpretation and say well you know if anything is a model really in the more traditional sense it's this recognition density that's performing posterior state and i'm sure that alex will tell me why i'm wrong in a second it's it's a great response let's just

move on through a few more of the slides because there's a lot in the figures that i think gets at this um but mel has their hand though sorry go ahead mel oh i just want to i just wanted to introduce it i think i think maxwell is possibly mismodeling himself i think he's i think he's actually a tad bit more realist than he's letting on oh maxwell a realist at heart on my bad days exactly if you can't if you can't uh take care of me at my most real so in this next slide one of the feedback just from a participant just to give a example and we won't spend too much time but i like this idea that the beliefs don't have to be verbal uh that the physical therapy context shows us how the range of movements that the body does not want to go to for example due to a traumatic experience or injury embodies a belief that doesn't involve language and this was a paper with uh limanowski and fristen from 2020 where a person's hand was opening and closing and then they were getting a vr visual that was delayed or was inaccurate and so there's so much here about how our sensory feedback are proprioceptive as well as our visual feedback multimodal systems are being used and that's the mess that would take us you know so many hours and more to clarify this is what the representation questions are about so it's like the person what is the representation of the current state of their hand being closed at that just pure phenotype level maybe somebody with a ruler says well it's whether the hand is closed or not but it's not just that because if you're getting visual input that's messing with you it changes your behavior so there's something else happening and that's a lot of what we're talking about um i think going through the figures two three four are going to be really helpful before we have closing thoughts and thanks everyone for bearing with this was sort of a weirdly technically altered stream briefly the bayesian networks so bayesian networks are from bayesian statistical sciences and in these diagrams the nodes are variables they're like random variables stochastic variables or parameters the edges are statistical relationships they're probabilistic relationships and they can be undirected or directed these models are quite common in statistics and also there's some really nice aspects about them in that you can bring prior knowledge to the table with your priors and you can also estimate it directly from the data with a parametric empirical base and to give an example about what this looks like here on the top right we can see how like an accident increases the likelihood that you'll hear sirens an accident also increases the likelihood of having a traffic jam bad weather it can increase the likelihood of accidents or traffic jams so these bayesian networks are pretty commonly used in a lot of statistical areas and we can contrast that with what are called forney factor graphs and working through this devries and first and paper was definitely quite an experience this is a complex paper and there's a lot in it but in a forney factor graph as they write in the paper it's a

type of graphical model that shares qualities with bayesian networks and markov random fields forney factor graphs afford a visually insightful representation of generative especially beneficial for complex models underlying hierarchical active inference in a forney factor graph each factor is represented by a node and each variable by an edge so it's like a flip of the bayesian an edge attaches to a node if the edge variable is an argument of the node and so here on the right is a table from the paper of devry's and fristen and it shows how we can think about these functions like addition subtraction multiplication as having variables coming in and out and just to really quickly summarize the similarities and differences in the bayesian networks the nodes are variables whereas in the forney graph the nodes are the factors or the functions in the bayesian networks the edges are the probabilistic relationships amongst whereas in the forney factor graphs the edges are the variables themselves a cool trick about bayesian networks is this model inversion which is that the model can be gleaned from the data and also the data can be generated from a model another cool trick of forney graphs is that they specify a specific order for factorizable computation making extremely massive models tractable and then a last cool fact about a bayesian network is that's where the markov blanket formalism comes from which is it's the set of nodes that insulates one part of the network from each other from a probabilistic perspective though there's a lot of nuance in how it's used in the free energy principle framework and the cool fact about the forney graphs is that it's linked to information theory and message passing computation and other math so in figure two we um see this generative model in active inference and this is actually the model that maxwell had on his screen earlier and i annotated all the pieces so just as maxwell was saying time flows from the left to the right and in this is a again a bayesian network representation so the the boxes or the circles and there's different types of them which are described in the caption the boxes and the circles are like variables so the initial state is a vector the hidden states are variables the state transition matrix is how the hidden states map to each other that's b a is the map between the states and the observations and then figure three is the same thing as a forney factor graph and so this is showing that there's this equivalence with representing something as a bayesian net and a forney factor graph and the forney factor graph allows for these extremely large scale hierarchical models to get factorized and then computed in a tractable way and so i should say like the advantage of using a forney style factor graph is that the message passing is just you can just read it directly off the forney style factor graph and there's an algorithmic way of moving from a bayes net to a forney style so typically intuitively the bayes net is easier to write uh because you can you know you can just literally write down

the dependencies between all of your factors like this uh so you know like i don't this state depends on this other state and you know there are these transitions and etc blah blah and so you know from this you can move to the forney style factor graph and that tells you just directly how that will be implemented in uh active inference i should say that the reason why i we discussed these at all is really just uh to show in the next figure how the in figure four how the generative model and the generative process kind of link up uh i mean to me this is where the the interesting stuff happens where like literally the the generative process and this is a model of the generative process right like how we think it's structured it's in this little box here um and then the 40 style factor graph represents the message passing that is ongoing yeah outside the box uh to keep all the the quantities within you know phenotypic bounds thanks maxwell and yeah i know that was a little bit of a rapid sprint through these figures but it's so cool that there's a formal way to relate the bayesian and the factor graph models and just as you specified it's easy to specify sometimes the bayesian network just like oh well the traffic jam depends on the weather and the weather depends on this other feature and then the factor graph allows us to have an order of computation and a really uh implementable way to compute it and also it has some similarities with message passing for example in neural network systems so it's it's an interesting question that i've been having is okay we talk about this is as a bayesian theory but if it can also be represented in another way is it is it really a uniquely bayesian theory or what do we gain by saying that it's a bayesian theory it's uh within the domain of bayesian inter-transformability uh or interoperability but that doesn't uniquely uh disambiguate it and that's why i think some of these philosophical questions that we're talking about here like about and about cause are so open because uh it's it's not uniquely specified by the free energy principle or by active inference like how exactly we should answer these philosophical questions because in the end as maxwell said it's the technical details this is what we're talking about when we're talking about the the free energy minimization and how that bears upon the policy selection this is what's happening and also one more point before if anyone raises their hand to provide a comment is um notice that the free energy is linking to the policy selection π p for policy and that means that the organism isn't just minimizing the free energy of the observations given a sort of neutral representation actually the policy representation specifically how it bears upon how states transition into each other that's what the actual computation is being done upon so the observations are being used to run then upstream through the hidden states and their transitions into the policy that ultimately dictates what's going to be done by being selected by the organism or the system as

the most likely so this is really how deeply action is embedded in these models it's not that there's a latent representation of the external states and then you're doing a best guess on what you should do it's actually like action is at the top here the policy is at the top and so i think that's really a nice feature of these models and it becomes graphically clear and then if somebody wants to have philosophical framework a or philosophical framework be on top of this structure or if they want to propose a different structure for how observations and policies are linked that's something great it's something tractable that we can talk about but this also returns to the metaphor discussion this is the real thing as it is um cool i know we had to sprint through there but it's just um coming near the end of our time so any thoughts on these images or are people ready to sort of go to our closing thoughts i was just saying the equations are more interesting but you have to say the equations are interesting and and the 2017 to verizon fristen and this model it just reminds uh me that if we can write down that bayesian relationship the probabilistic relationship between variables then there's going to be a process model that is going to be how we can link it and these captions are very informative and there's really a lot to say here but enough for today let's close it out um we can hear any last thoughts but while people are preparing their last thoughts and raising their hand if they'd like i would just like to thank everyone for participating we're learning and developing each time figuring out the tech each time we provide follow-up forms to the live participants so just check the calendar invitation and it would be really helpful to have we also request feedback suggestions and questions from the participants live in the comments or otherwise and uh just stay in communication we really appreciate everyone's engagement here and it's been such a great learning experience so if anyone wants to raise their hand alejandra first and then followed up by anyone else who wants to let's just have some closing thoughts well i was wondering if it's possible to like model updating if when you are doing offline cognition um we like without like an explicit policy you are not doing something online when this interaction bodily interaction you are thinking about something and then you have like this uh like you know said like this eureka feeling so this this is kind of a model updating um without action um i don't know the short answer is yes i just linked the paper to it we uh we've created a ruminating agent um so basically this agent combines the affective inference uh scheme that we uh have developed over the last two years where you're effectively inducing a new hierarchical layer of state inference based on state and precision estimates at the lower level and these are affective states that then guide lower level precision estimation so we combine that with what's called sophisticated inference which is the new generation of active inference

agents um you know vanilla formulation current vanilla um you could see that in the generative models that we just showed they include uh beliefs about counterfactual observations that we would make as we propagate states into the future sophisticated inference is about propagating your beliefs about the observations that you would make in counterfactual futures and so by combining these two things we have an agent that isn't acting uh but that ends up ruminating um and catastrophizing about uh its own cognition um so i just linked the paper in the chat um we we submitted it to the uh the international workshop on active inference uh so the it's a little poster now but we're we're slowly expanding that into a full manuscript but the the kind of summary six pager uh that just does what you were suggesting ale is uh this is it yeah thank you maxwell um but this is kind of confusing with the an active uh view of cognition well this is why we're not we're not like uh classical and activists so yeah we reject the equal partners principle it's just not the case that uh the body uh the brain and the cultural like mealier are going to be involved in the same way for every task so like for rumination you know you could you could i guess you know appeal to a history of interaction which is you know the reason why agents are able to entertain beliefs about other agents at all and so on and so forth but yeah you're right there isn't much in action here and that's fine i think alex kiefer and then anyone else who wants to raise their hand uh yeah so i guess in lieu of a proper statement i just wanted to point out that maxwell has some really cool new work on how the neuronal populations sort of can be seen as an acting uh inference on their own level and that this is one way you could see uh structural representations being sort of realized so like that one thing we didn't have time to get into um but i just wanted to say in general like i think um yeah there is i think regardless of the precise philosophical take that you have on all this stuff like it's cool to disagree on that level but this is this literature has produced some really amazing work and i think it's it's pushing forward in a really interesting way and one one advantage of active difference certainly is that you can actually write down the generative model and and sort of transparently see how it works and you don't get that out of just unsupervised learning who any other thoughts awesome well thanks so much for the discussion we're gonna um re-watch understand technologically what did didn't work and for next week provide all of you plus anyone else who wants to participate with more of a specific spec sheet for how you can prepare and make sure that it will be a really like seamless experience and as far as the timing goes as well next week we will move on to active seven which is going to be about variational ecology which is a really awesome paper close to my heart as a biologist as well and we would really appreciate anyone who wants to participate in

these conversations next tuesday at 7 a.m pacific or the following tuesday both of these discussions will be really great and it's quite a complex paper so i'm sure we'll have a lot to go into again so once again thanks so much everyone for your participation really brought up some awesome ideas and i hope we conveyed to those within and without of the field that there's a lot of discussion ongoing by no means are these questions resolved and it's an opportunity for people to bring their unique perspectives to the table so thank you all again for your really appreciated and we'll see you later